

MICHAEL GURULE

Applied Scientist and ML Architect

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SUMMARY

Applied scientist and ML architect who builds statistical models and the cloud systems that run them, mostly on operational data that arrives noisy, sparsely labeled, and changing. Currently own the modeling, cloud architecture, and technical roadmap for a greenfield anomaly detection, correlation, and forecasting platform covering about 32 million broadband endpoints. Nine years of technical consulting across financial services, industrial commodities, and telecommunications before that, with an MS in Applied Mathematics underneath the work.

EXPERIENCE

Principal Engineer II, Telemetry and AI Operational Intelligence

Oct 2025 – Present

Spectrum (Charter Communications)

Applied science and ML architecture lead for the development of an anomaly detection, correlation, and forecasting platform

- Designed the AWS architecture for the platform, from streaming ingestion through inference, retraining, and monitoring, and oversaw the move of telemetry and ML workflows from on-premises systems to a hybrid cloud.
- Built a custom ensemble model for network telemetry by studying the source of Prophet, NeuralProphet, and AutoGluon and reimplementing only the methods that held up on this data; adapted rules-based models first to set the baseline it has to beat.
- Score each device against peer groups of comparable devices in comparable operating states, which keeps detection sensitive to local faults without fitting a separate model for each of 32 million devices.
- Separated deterministic threshold alerts from statistical detection, running Random Cut Forest and seasonal residual analysis on a five-minute cycle so normal daily load swings stop registering as faults.
- Reduced feature dimensionality to fit the full fleet inside each five-minute scoring window while keeping the signals that separate real faults from noise.
- Own model release decisions: define SLAs and performance thresholds, set training cadence and data windows to match each method's statistical assumptions, and decide whether a model goes to production, waits, or gets reworked.
- Designed the labeling, ground truth, and benchmark strategy for model evaluation, and use production diagnostics to set ingestion, cleaning, and validation requirements for the data pipeline.
- Found that the inherited rules platform capped detection accuracy at the level of its own rule outputs; designed a pipeline that reads raw telemetry directly and keeps full signal resolution.
- Building cross-topology and hardware correlation with confidence weights calibrated from operator feedback, turning separate alerts into ranked root-cause candidates.
- Wrote the platform roadmap with the Director of Operational Intelligence, from proof of concept through correlation, forecasting, and expansion to new metric groups and business units; now defining scope and guardrails for agents that act on detected anomalies.

Senior Data Scientist and Machine Learning Engineer

Mar 2020 – Aug 2025

Sedgwick

- Built a serverless AWS Lambda pipeline that automated risk scoring across more than 3,000 municipal jurisdictions, with version control for both data and model priors.
- Built Bayesian neural networks, LDA topic models, and Gaussian mixture models in PyMC and TensorFlow for marketing attribution and client segmentation.
- Measured incremental marketing effect with Bayesian structural time series and identified profit drivers with multivariate regression, contributing to a six-figure EBITDA improvement.
- Set up drift monitoring with Kolmogorov-Smirnov tests and Population Stability Index, tied to automatic retraining, so models held up through regulatory changes.
- Consolidated ERP systems and public sector APIs into one AWS data lake, cutting data latency for portfolio risk monitoring by 20%.

- Built time series and regression models for more than 13 advisory engagements, forecasting multimillion-dollar financial and operational trends for industrial and financial clients.
- Mapped legacy data schemas to AWS and relational SQL infrastructure during a major M&A integration, and led the pipeline assessments that set the scope of the cloud migration.

SELECTED PROJECTS

Source and implementation on GitHub.

SENTINEL: Multi-Sensor Positioning. Fuses RF positioning (TDOA/FDOA), thermal imaging, and telemetry with an extended Kalman filter for real-time tracking across asynchronous streams. Better sensor placement geometry raised tracking accuracy 40-fold in GPS-denied conditions with no change to the estimator; the gain came from improving how well the problem was conditioned.

CONSTELLATION: Satellite Fleet Health. Anomaly detection for LEO constellations using an Isolation Forest and LSTM autoencoder ensemble, with Temporal Fusion Transformers for forecasting and Cox proportional hazards for time to failure. Designed around sparse labels and seasonal load.

MERIDIAN: Portfolio Optimization Under Regime Shifts. Regime detection, covariance estimation, and GARCH(1,1) return forecasts feeding a portfolio optimizer with realistic transaction costs, across 25+ assets and ten years of data.

HYPERION: Multi-Agent Reinforcement Learning. Coordinates UAV swarms with Ray RLLib and PyTorch, using physics-informed neural networks for trajectory simulation and graph neural networks for agent communication. Sub-100ms inference and an 85% interception rate with more than 20 agents.

TECHNICAL

Statistical Modeling: Regularized regression, GAMs and splines, local regression, Bayesian and hierarchical models, Bayesian structural time series, survival analysis, change-point detection, Kalman filtering and state estimation, hypothesis testing

Machine Learning: Gradient boosting, random forests, Isolation Forest, Random Cut Forest, LSTM autoencoders, Temporal Fusion Transformers, Gaussian mixture models, reinforcement learning, graph neural networks, physics-informed neural networks

Evaluation and Model Ops: Feature engineering, regularization, model selection, cross-validation for time-dependent data, probability calibration, drift detection, hyperparameter tuning, experiment tracking, model versioning

Mathematics: Convex and non-convex optimization, information theory, estimation bounds, Fisher information, numerical linear algebra

Cloud and Systems: AWS (SageMaker, Bedrock, MSK, Managed Apache Flink, S3, Glue, Athena, Neptune, EventBridge), Kafka, Flink, Druid, Spark, Docker, Kubernetes, CI/CD, FastAPI

Languages: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, TensorFlow, PyMC, CVXPY), SQL (PostgreSQL, MySQL), JavaScript

EDUCATION AND CREDENTIALS

MS Applied Mathematics, University of Colorado	2024
BS Finance and Risk, University of Colorado Denver	2017
AWS Certified Machine Learning Engineer, Associate	